Shortest Distances in 3D: Points and Lines

Questions 1–33

Question 1. Find the shortest distance from the point $(2, 5, -3)$ to the x-axis.

Question 2. Find the shortest distance from the point $(1, 8, -9)$ to the z-axis.

Question 3. The point P has position vector $\overrightarrow{OP} = 6\mathbf{i} + 12\mathbf{j} - 5\mathbf{k}$. The line l passes through the point P and is perpendicular to the vector $\overrightarrow{OP}$. Find the shortest distance between l and the origin O.

Question 4. Find the shortest distance from the origin to the line with equation $\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$.

Question 5. Find the shortest distance from the point $(2, 1, 1)$ to the line with equation $\mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$.

Question 6. A sphere has centre $(-4, 3, 8)$ and radius r. Its surface touches the line that passes through $(1, 0, 1)$, parallel to the vector $\mathbf{i} + \mathbf{j}$. Find r.

Question 7. Find the shortest distance from $(1, 0, 1)$ to the line with equation $\frac{x-1}{2} = y = z + 2$.

Question 8. Find the shortest distance from $(6, 0, 4)$ to the line with equation $x = 2y = 3 - z$.

Question 9. It is given that the points A and B have coordinates $(2, -1, 1)$ and $(9, 12, 4)$ respectively. The line l passes through the points A and B. The point P has coordinates $(9, -22, 0)$.

a) Show that P does not lie on the line l.

b) It is given that Q is the point on l that is closest to point P. Find the coordinates of the point Q.

c) Hence find the exact shortest distance between the point P and the line l.

Question 10. Given that the points $A(3, -1, 0)$ and $B(5, 1, 2)$ lie on the plane $\Pi : 2x + 7y - 9z = -1$, find the equation of the line that lies on the plane Π such that every point on the line is equidistant from A and B.

Question 11. The points P and Q have coordinates $(2, 2, 8)$ and $(10, 3, 0)$ respectively.

a) Show that P and Q lie on the plane Π with equation $3x - 8y + 2z - 6 = 0$.

b) Find the equation of the line l that lies on the plane Π such that every point on the line is equidistant from P and Q.

c) The point R lies on the line l such that the chord $PR = \frac{43}{\sqrt{6}}$ units. Find the possible coordinates of R.

Question 12. The points A, B and C have coordinates $(2, 1, 2)$, $(3, 4, 1)$ and $(0, 1, 1)$ respectively.

a) Find the cartesian equation of the plane Π that contains A, B and C.

b) Find the equation of the line l_1 contained by Π such that every point on l_1 is equidistant from A and B.

c) Find the equation of the line l_2 contained by Π such that every point on l_2 is equidistant from B and C.

d) It is given that P is the point of intersection of l_1 and l_2. Find the coordinates of P.

Question 13. An underground railway tunnel can be modelled by a straight line given by the equation

$$\mathbf{r} = \begin{pmatrix} 50 \\ -765 \\ -116 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix}.$$

The ground above can be modelled by a plane given by the equation $x - y + 6z = 300$. An emergency exit from the tunnel, P, is to be built to connect to a point $Q(10, k, 10)$ on the ground.

a) Find the value of k.
b) Find the coordinates of P such that the distance from Q to P is minimised.
c) Hence find the exact distance from Q to P.

Question 14. It is given that the line l has equation $\mathbf{r} = \mathbf{i} + \mathbf{j} + 2\mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} + \mathbf{k})$ and the point P has coordinates $(k, 2, -1)$.

a) Show that the shortest distance between P and the line l, in terms of k, is $\frac{1}{3}\sqrt{6k^2 + 12k + 24}$.
b) Hence find the value of k that gives the minimum value of $\frac{1}{3}\sqrt{6k^2 + 12k + 24}$. State this minimum value.

Question 15. The point A has coordinates $(1, 2, 1)$. Find the coordinates of the image of the point A after it is reflected in the line with equation $\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$.

Question 16. It is given that the point A has coordinates $\left(29, 29, -\frac{109}{3}\right)$. Find the coordinates of the image of the point A after it is reflected in the line with equation $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix}$.

Question 17. Find the shortest distance between the following pair of parallel lines:

$$l_1 : \ \mathbf{r} = \mathbf{i} + \lambda(\mathbf{j} + \mathbf{k}), \qquad l_2 : \ \mathbf{r} = \mathbf{k} + \mu(\mathbf{j} + \mathbf{k}).$$

Question 18. Find the shortest distance between the following pair of parallel lines:

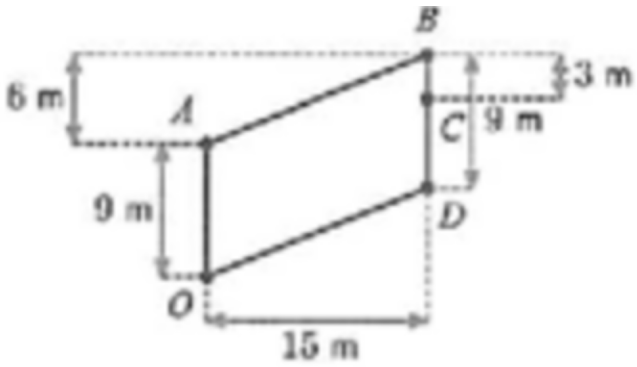
$$l_1 : \ \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \qquad l_2 : \ \mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

Question 19. Find the shortest distance between the lines with equations:

$$l_1 : \ \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \qquad l_2 : \ \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}.$$

Question 20. Two crane arms are to be used in a movie stunt. The crane arms are modelled as straight lines using coordinates in metres. One crane arm has its base at $(20, 20, 60)$ and its end at $(80, 80, 90)$. The other crane arm has its base at $(60, 20, 70)$ and its end at $(30, 70, 100)$. Find the shortest distance between the two crane arms.

Question 21. A drone light show involves two drones that are programmed to follow straight paths at different angles. The drones are programmed to remain within the parallelogram shown in the diagram below, as seen from above. Take the point O to have coordinates $(0, 0, 0)$. Drone 1 starts from 18 metres above point A and follows a straight path to reach a height of 4 metres above point D. Meanwhile, Drone 2 starts from 3 metres above point B and follows a straight path to reach a height of 4 metres above point O, before then immediately changing direction and following a straight path to reach a height of 20 metres above point C.



Determine whether the flight path for Drone 2 is closer to the flight path of Drone 1 when it is travelling from above B to above D, or when it is travelling from above O to above C.

Question 22. Two ski lifts at a resort are to be built so that they cross over one another. The first is to pass between the two points $(-180, -20, 25)$ and $(260, 2, 36)$. The second is to pass between the two points $(-220, 44, -1)$ and $(290, 10, 50)$. All measurements are given in metres, where the origin is the reception chalet. For safety, the two ski lifts cannot pass within 10 metres of one another.

a) Determine whether the design of the two ski lifts passes the safety regulation.

b) State any assumptions that have been made that may affect the design passing the safety regulations.

Question 23. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} k \\ 2 \\ 1 \end{pmatrix}, \qquad l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix},$$

where k is a constant. It is also given that the shortest distance between the lines l_1 and l_2 is 2 units. Find the value of k.

Question 24. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ k \\ k \end{pmatrix}, \qquad l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix},$$

where k is a constant. It is also given that the shortest distance between the lines l_1 and l_2 is 1 unit. Find the possible values of k.

Question 25. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} -1 \\ k \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} k \\ 2 \\ -1 \end{pmatrix}, \qquad l_2 : \mathbf{r} = \begin{pmatrix} k \\ k \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 3k \\ -2 \\ 2 \end{pmatrix},$$

where k is a positive constant. It is also given that the shortest distance between the lines l_1 and l_2 is 1 unit. Find the value of k.

Question 26. Two cables are hung across a room near the ceiling. The cables are modelled as straight lines using coordinates in metres, relative to a fixed origin O. One cable runs from the point $(3.5, k, 2.5)$ to the point $(1.5, 0.5, 3)$. The other cable runs from the point $(1.5, 1, 3)$ to the point $(4, 1.5, 2.5)$. The two cables are to be attached at their closest point by a wire that is 40 cm in length. Find the possible values of k.

Question 27. Find the shortest distance between the lines with equations, giving your answer to 3 significant figures:

$$l_1: \ \frac{x+2}{4} = \frac{y-5}{2} = \frac{z+7}{3}, \qquad l_2: \ \frac{x-1}{8} = \frac{y+2}{4} = \frac{z-4}{5}.$$

Question 28. Find the exact shortest distance between the lines with equations:

$$l_1: \ x-1 = \frac{y+3}{5} = \frac{z}{2}, \qquad l_2: \ \frac{x+5}{2} = y-3 = 1-z.$$

Question 29. Find the exact shortest distance between the lines with equations:

$$l_1 : \ 2x + 5 = 4 - y = 5 - 2z, \qquad l_2 : \ \frac{x}{2} = 4y = z - 3.$$

Question 30. It is given that the shortest distance between the following lines is $\sqrt{2}$ units:

$$l_1 : \ x - 1 = 2 - y = 4 - z, \qquad l_2 : \ x = 1, \ ky = z,$$

where k is a constant. Find the value of k.

Question 31. It is given that two lines have equations:

$$l_1 : \left(\mathbf{r} - \begin{pmatrix} 7 \\ 6 \\ 10 \end{pmatrix}\right) \times \begin{pmatrix} -6 \\ 5 \\ 3 \end{pmatrix} = \mathbf{0}, \qquad l_2 : \left(\mathbf{r} - \begin{pmatrix} 0 \\ 2 \\ -9 \end{pmatrix}\right) \times \begin{pmatrix} 6 \\ -5 \\ -3 \end{pmatrix} = \mathbf{0}.$$

a) Explain how you know that the two lines are parallel.
b) Find the exact distance between the two lines.

Question 32. Find the exact shortest distance between the lines with equations:

$$l_1 : \left(\mathbf{r} - \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}\right) \times \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} = \mathbf{0}, \qquad l_2 : \left(\mathbf{r} - \begin{pmatrix} -1 \\ 5 \\ -3 \end{pmatrix}\right) \times \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix} = \mathbf{0}.$$

Question 33. Find the exact shortest distance between the lines with equations:

$$l_1 : \ 2x + 3 = 3 - z, \ y = 2, \qquad l_2 : \ \left(\mathbf{r} - \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}\right) \times \begin{pmatrix} -7 \\ 2 \\ 3 \end{pmatrix} = \mathbf{0}.$$